

Margaux Delizée

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Education

M.Eng. - Biomedical Engineering | Boston University, USA | Sept. 2026–May 2027

- BAEF Fellow (Belgian American Educational Foundation) & BU Dean's Graduate Scholarship recipient

M.S. – Electrical Engineering | University of Mons, Belgium | Sept. 2024–Jun. 2026

- Average: 18/20 (*Summa cum laude*).
- Specialization in Life Sciences Applications.
- Relevant coursework: Robotics and Biomedical Applications, Medical Image and Signal Processing.

Exchange program – Electrical Engineering | Polytechnique Montréal, Canada | Aug. 2024–Dec. 2024

- 3.9 GPA
- Relevant coursework: Signal Analysis, Multivariable Systems, Biomedical Measurements.

B.S. – Electrical Engineering | University of Mons, Belgium | Sept. 2021–Jun. 2024

- Graduated with honors, top 10%.
- Relevant coursework: Physics, Thermodynamics, Electromagnetism, Electronics, Differential Equations, Probability and Statistics, Calculus, Physical and Organic Chemistry.

High School Diploma | INDBE, Braine-le-Comte, Belgium | Sept. 2015–Jun. 2021

- Focus on Mathematics and Sciences.

Research & Academic Projects

Master's thesis | Lehigh University PA | Feb. 2026–Jun. 2026

- Developed closed-loop control algorithms for cardiovascular dynamics regulation via vagal nerve stimulation, targeting atrial fibrillation in a human in silico model.
- Submitted abstract for poster presentation at SfN Annual Meeting, Nov. 2026 (under review).
- Manuscript in preparation for journal submission.

Control Strategies for Drug Delivery | University of Mons | Feb. 2025–Jun. 2025

- Designed advanced drug delivery controllers in a surgical patient simulator (MATLAB).
- Independently authored a research article exploring model-based control approaches.

ECG Acquisition & Analysis | Polytechnique Montréal | Sept. 2024–Dec. 2024

- Recorded and analyzed ECG signals using Biopac.
- Applied filtering and signal processing techniques to detect cardiac abnormalities.

Smart Activity Monitoring Device | University of Mons | Feb. 2024–Jun. 2024

- Built a wearable fitness tracker with heart rate, accelerometer, and GPS sensors (with Arduino).

Automated Object Sorting System | University of Mons | Sept. 2021–Dec. 2021

- Designed and implemented an automated sorting system integrating hardware and software design.

Professional Experience

System Engineer Intern | IBA, Louvain-la-Neuve, Belgium | Jul. 2025–Oct. 2025

- Contributed to the development of control algorithms for the proton therapy gantry system, enhancing the precision of treatment with the proton beam.
- Collaborated with engineers to integrate control, software, and biomedical requirements.
- Performed 3-day on-site testing of control algorithms at a proton therapy center in Reading, UK.

Mathematics Tutor | Jan. 2022 – Present

- Provide mathematics tutoring for high school and university students, including preparation for engineering entrance exams, with a focus on algebra, calculus, and geometry.

Skills

Programming: Python (NumPy, PyTorch), C++, Arduino, MATLAB (Simulink) • Biomedical signal processing (filtering, feature extraction) • Scientific writing & publication • Computational modelling • Interdisciplinary teamwork • Project leadership • French (native), English (C1, TOEFL 103/120).